

Predictive filtering-based fast reacquisition approach for space-borne acquisition, tracking, and pointing systems

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Abstract: We propose a novel approach for recapturing targets as quickly as possible after the space-borne ATP (Acquisition, Tracking, and Pointing) system target loss in free-space laser communication. The approach uses past angular information to predict the target trajectory. An optimized finite memory filter is designed as the prediction algorithm. To obtain optimal prediction performance, the designed filter determines the filter parameters of different curve parts during offline training, and the parameters are adjusted according to the curve characteristics during predictive filtering. Simulation results indicate prediction accuracy above 0.1 degrees within 5 seconds. An experimental system was constructed in the lab to simulate the reacquisition process using a dynamic target and a real ATP system. Experimental results show that, compared with the classical orbit prediction method, this approach can effectively yield shorter acquisition times.

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1. Introduction

High capacity, high transmission rate, and excellent security make free-space laser communication a promising prospective communication technology. Since 1994, countries and institutions including Japan and ESA had successfully performed several experiments demonstrating satellite-to-ground and satellite-to-satellite laser communication [1–3]. In June of 2014, NASA transmitted a HD video "Hello World" from the International Space Station to the ground station by using a laser beam in its optical communication project OPALS [4]. Accumulating evidence indicates that free-space laser communication is being used extensively in a growing number of areas. This increasing popularity requires increasing the robustness of the ATP (Acquisition, Tracking, and Pointing) system which is used to establish high accuracy, high stability optical tracking-and-pointing link.

As applied to communication terminals, an important aspect of the ATP system's robustness is the ability to quickly recover following optical link interruption. For laser communications between LEO satellites and ground stations, limited by visual segments, the communication time of each access is usually a few hundred seconds. Figure 1 shows the duration of visual access of a LEO satellite-to-ground link in a time window of one month. Figure 2 shows schematic of a typical laser communication workflow during one pass. During the tracking process, satellite-to-ground optical link may be interrupted by optical signal fading or even by completely blocked LOS (line of sight, hereinafter referred to as LOS), due to atmospheric conditions such as turbulence, clouds, and rain [5] (Fig. 2), requiring link recovery. As mentioned above, because the total communication time is short and the data rate is high, even tens of seconds of communication interruption can significantly affect the average data rate and the total data volume [5]. Obviously, link recovery time should be as short as possible, which requires the ATP system - mainly the acquisition subsystem - to quickly reacquire the target and establish the link.

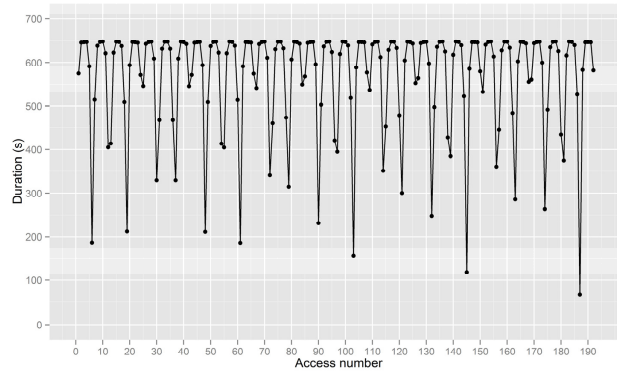


Fig. 1. Visual access duration during one month, for a LEO (altitude = 400 km) satellite-to-ground link simulated in STK.

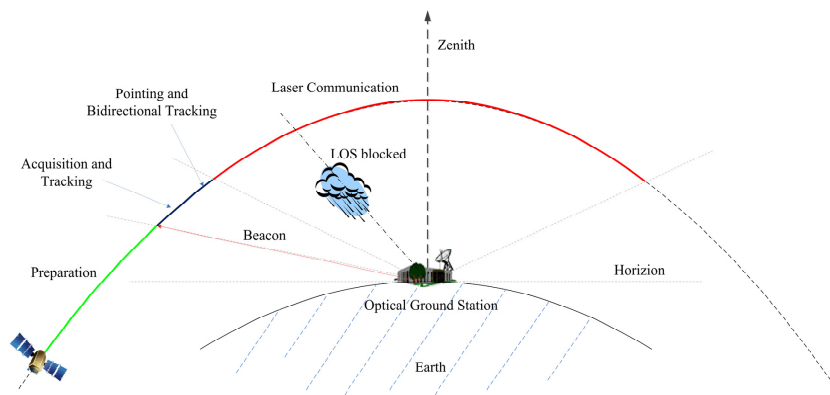


Fig. 2. Schematic of laser communication in a LEO satellite pass over the ground station.

A typical acquisition process comprises three procedures: search, recognition, and execution. During the search step, the FOU (Field of Uncertainty, hereinafter referred to as FOU, mainly caused by the inaccuracy of orbit prediction and satellite attitude measurement) of a target is searched. Next, the light spot in the FOV (Field of View, hereinafter referred to as FOV) is recognized to confirm the communication optical signal. Finally, an actuator is driven by a controller to position the target optical axis at the FOV center. Accordingly, the operation time of these three procedures constitutes the total acquisition time. *Tan et al.* [6] and *Zhang et al.* [7] studied quick recognition of optical signals and moving characteristics of driving motors, respectively, and they demonstrated that recognition time and execution time both contribute no more than 100 ms to the acquisition time. In fact, most of the acquisition time is devoted to the search procedure. For example, the maximal acquisition time of STROPEX space-to-ground optical communication is 30 s [8] while for Artemis satellite-to-ground optical communication the acquisition time is 51 s [9]. In order to shorten the scanning phase of the acquisition process, scanning strategy optimization techniques have been extensively studied [10–13]; however, only a limited improvement of acquisition time has been achieved, because even an optimized scanning process still takes tens of seconds.

Another way to shorten the reacquisition time is by using prediction methods. Assuming that the optical link interruption does not necessarily affect the relative motion of the target beam, it might be possible to use the historical information for inferring the target beam trajectory and to use the trajectory for open-loop pointing. By using the platform vibration data and by assuming a constant change rate of the acceleration in a short period of time, *Lee and Ortiz* [5] predicted the beacon movements on the camera when the signal faded (or was blocked), and the maximal prediction time was 3 seconds (corresponding to a prediction error

of 0.07°). This approach is helpful for fast link reestablishment, but it relies on the inertial sensors' platform data, and the prediction time is relatively short.

This paper presents a new approach toward fast reacquisition, based on predictive filtering. During the “dark period”, while the space-borne ATP system's target optical signal is not available, the historical information measured within the system, such as angle data of the gimbal and miss distance data of the tracking camera, is used to predict and extrapolate the target position. The predicted position is used for pointing, and with a reduced FOU, the search time is shortened, achieving fast reacquisition. Here, we address the following three issues:

1. Analyzing the prediction algorithm feasibility based on the angular information.
2. Designing an effective prediction algorithm by considering the pointing angle characteristics.
3. Verifying the fast reacquisition approach results obtained by using the designed algorithm.

2. The principle of predictive filtering

2.1 Predictive filtering based on angular information

In general, two types of methods exist for predicting the two-axis gimbal pointing angle of the space-borne ATP system during the dark period: 1) predicting in the Cartesian coordinate system, and then calculating the pointing angle through coordinates transformation, 2) predicting the angle directly in the spherical coordinate system [14]. Because the first method requires the range information between the satellite and the ground station, and because the space-borne ATP system usually has no access to precise range data, the second method, which only requires angle information of the ATP system, is adopted for predicting. As shown in Fig. 3, the LEO satellite's pointing angle to the ground station, as well as its first and second time derivatives, nonlinearly depends on time and is model-unknown; thus, the predictive filtering algorithm's feasibility should be analyzed.

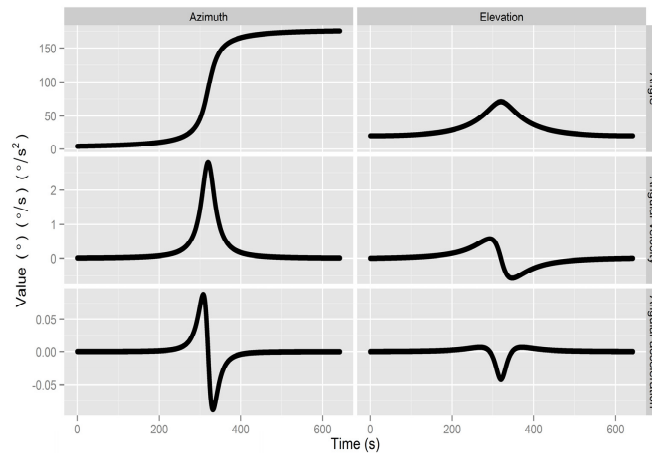


Fig. 3. Pointing angle to the ground station of a LEO satellite (altitude = 400 km), along with its first and second time derivatives, simulated in STK.

Predictive filtering is based on mining useful information from noisy historical data, thereby extracting the motion model, and using this information to extrapolate the future trajectory. Therefore, adequate and effective historical information is necessary for the predictive algorithm. As for the space-borne ATP system, during the tracking process, the optoelectronic gimbal records the high-precision angle data and miss distance data from its

encoder and camera, respectively, and the accuracy of these internal measurements is much higher than the external data such as orbit prediction. After aligning and combining these data the target's real pointing angle can be obtained, which contains useful information such as the satellite's relative motion. Based on the motion continuity, the approaching angle can be deduced for a short period.

In addition, the movement mobility directly affects the prediction results. For the three-axis stabilized satellite platform, the angular movement relative to the target is mainly caused by the satellite's motion around the Earth, which is relatively stable. Yet, it should be noted that the satellite platform vibration introduces an additional high-frequency noise [15,16]. Furthermore, for each passage of the LEO satellite over the ground station the pointing angle movement is different, which is related to $E_{\max}$, the maximal elevation angle of the ground station. In this paper, we studied the LEO satellites at the altitude of 400 km. Table 1 shows the angular movement parameters of three different trajectories corresponding to three passes over the station. As is apparent from Table 1, the angular movement mobility increases with increasing $E_{\max}$, and the azimuthal angle mobility is remarkably higher than that of the elevation, which corresponds to the high-speed movement along the satellite orbit.

Table 1. Angular Movement Parameters of a LEO Satellite (Altitude = 400 km) Pointing to Ground with Respect to the Ground Station's Maximal Elevation Angle $E_{\max}$

	$E_{\max} = 40^\circ$ (Trajectory 1)	$E_{\max} = 55^\circ$ (Trajectory 2)	$E_{\max} = 70^\circ$ (Trajectory 3)
Maximal motion parameters			
Angular velocity of azimuth	$0.967^\circ/s$	$1.515^\circ/s$	$2.790^\circ/s$
Angular acceleration of azimuth	$0.011^\circ/s^2$	$0.026^\circ/s^2$	$0.088^\circ/s^2$
Angular velocity of elevation	$0.207^\circ/s$	$0.356^\circ/s$	$0.567^\circ/s$
Angular acceleration of elevation	$0.008^\circ/s^2$	$0.018^\circ/s^2$	$0.042^\circ/s^2$

2.2 N-points finite memory filter

The key component of predicting is an appropriate predictive filtering algorithm. The algorithms commonly used in optoelectronic tracking systems include α - β - γ filters, Kalman filters and N-points finite memory filters [17]. These are often used to improve the tracking accuracy [18], forecast the radar-pointing angle [19], and predict the target trajectory [20]. In this paper, considering the fact that the angular motion model of the space-borne ATP system is unknown, we adopted the N-points finite memory filter as the basic prediction algorithm, while the other two kinds of filters could be alternatives [21,22]. Next, the basic concept of N-points finite memory filter will be briefly introduced.

The N-points finite memory filter uses the latest N points of data, which contain measurement noise, to estimate the current and future data, ensuring a minimal mean squared error [17]. This filter is also known as the polynomial filter, because polynomial curve fitting is often used in this method [23].

Let us assume that the N-points historical measurement data is $y_1, y_2 \dots y_N$, and the fitting curve is an m-order polynomial P_k :

$$P_k = \sum_{j=0}^m a_j k^j, k = 1, 2 \dots N. \quad (1)$$

The deviation of the fitting data from the measured data is $\delta_k = P_k - y_k$. A set of coefficients, a_j , has to be determined that minimizes the sum of the squared errors:

$$\sigma = \sum_{k=1}^N \delta_k^2 = \sum_{k=1}^N \left(\sum_{j=0}^m a_j k^j - y_k \right)^2 = \min. \quad (2)$$

After obtaining the fitting parameters by using the least squares method, extrapolation can be performed by using the fitting function P_k .

3. Design and simulation of the optimized finite memory filter

3.1 The finite memory filter problem

The number of memory points and the function class are key determinants of finite memory filters prediction performance. Using more points confers stronger ability to filter the random disturbances; however, the effect of new data will be weakened, resulting in a reduced ability to respond to mobility. Using fewer points increases the randomness and may cause instability [17]. The function class determines the fitting curve type; for polynomial fitting, it mainly refers to the polynomial order. Using higher-order polynomials can yield more flexural curves; however, an improperly high order may result in extra error for trajectories close to straight lines. Further, high order polynomials increase the calculation complexity, and sometimes yield matrices with ill-conditioned coefficients. Thus, the selection of a proper order should be based on the curve characteristics.

As shown in Fig. 3, the pointing angle curve from the LEO satellite to a ground station has unique features. The lowest angular mobility in the curve appears at both ends because of the longest range; the mobility is much higher in the middle, where the angular velocity and acceleration attain larger values. By using simulations, we found that for this angle curve type, there is a problem of parameter adaptability for the finite memory filter. That is, if a fixed value parameter is applied, it is difficult to ensure the prediction effects on different curve parts. To illustrate this let us consider the following example. Let us assume there are two different sets of filter parameters, $P1$ and $P2$:

$$P1 = \{N = 675, m = 2\}. \quad (3)$$

$$P2 = \{N = 475, m = 5\}. \quad (4)$$

where N stands for the number of memory points used for fitting, and m denotes the fitting function order.

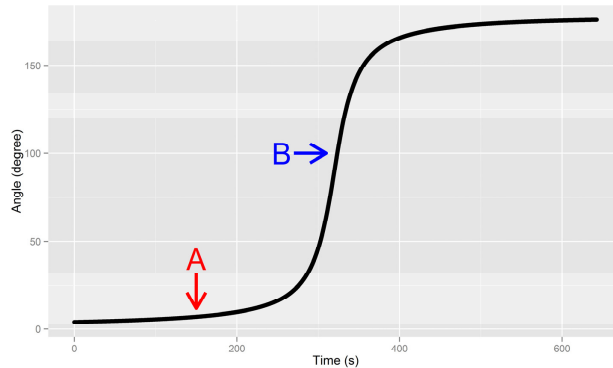


Fig. 4. Two selected prediction points on the curve of azimuthal angle (Point A: time = 150 s, Point B: time = 320 s).

Select two points, point A and point B, on the azimuthal angle curve in Fig. 3 as the two prediction points. As shown in Fig. 4, the angular velocity at point A is $0.04^\circ/s$, and that at point B is $2.80^\circ/s$. The 5 seconds long prediction calculation is then performed at points A and B using the parameter sets $P1$ and $P2$.

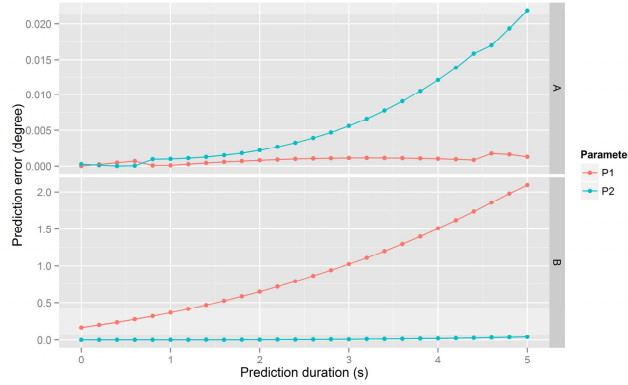


Fig. 5. Prediction results of two different sets of filter parameters at points A and B.

As can be seen from the prediction results in Fig. 5, there are some contradictions at the two points A and B. Namely, if the parameter set $P1$ is used at both points, point A will yield a good prediction, but the prediction error at point B will increase to an unacceptable level. Conversely, if parameter set $P2$ is used, a better prediction will be obtained at point B, but the prediction error will significantly increase at point A. In addition, while the smaller number of memory points in $P2$ may cause instability, the unnecessarily high polynomial order dramatically increases the computation time (see specific calculation in Eq. (6)). In other words, in classical finite memory filters with fixed parameters there is a tradeoff between stability and randomness, and between accuracy and computational complexity. Therefore, we have designed an optimized finite memory filter to achieve best filter performance on different curve parts.

3.2 Design of the optimized finite memory filter

Apparently, an optimized algorithm should take into account prediction accuracy and computational complexity when selecting the number of memory points N and the fitting order m . Prediction accuracy can be assessed by the 5 seconds long prediction error calculation, which we denote $Error_{5s}$. Computational complexity is analyzed by solving for the fitting coefficients. To solve Eq. (2), we start with the following equation:

$$\begin{bmatrix} N & \sum_{k=1}^N k & \cdots & \sum_{k=1}^N k^m \\ \sum_{k=1}^N k & \sum_{k=1}^N k^2 & \cdots & \sum_{k=1}^N k^{m+1} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{k=1}^N k^m & \sum_{k=1}^N k^{m+1} & \cdots & \sum_{k=1}^N k^{2m} \end{bmatrix} \times \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_m \end{bmatrix} = \begin{bmatrix} \sum_{k=1}^N y_k \\ \sum_{k=1}^N k y_k \\ \vdots \\ \sum_{k=1}^N k^m y_k \end{bmatrix}. \quad (5)$$

Writing Eq. (5) as $XA=Y$ the polynomial fitting coefficients are obtained as $A=X^{-1}Y$. Thus, the computational complexity consists of: constructing the matrix X (which takes $N(2m-1)$ multiplication operations), calculating the inverse of X (with associated complexity of $O(m^3)$ [24]), constructing the matrix Y (which requires $N*m$ multiplication operations), calculating the matrix $X^{-1}Y$ (with associated complexity of $O(m^2)$). Thus, overall computational complexity is:

$$T = N(3m-1) + O(m^3) + O(m^2). \quad (6)$$

Because the polynomial order m is usually not higher than 5, and because the number of the fitting points N is about several hundred (of course this is determined by the sampling interval), the second and the third terms in Eq. (6) can be neglected, so that:

$$T = N(3m-1). \quad (7)$$

Next, we use the prediction error $Error_5s$ and the computational complexity T to construct the penalty function F , which is a function of N and m :

$$F(N, m) = Error_5s + \frac{N(3m-1)}{N_{\max}(3m_{\max}-1)} * K. \quad (8)$$

The computational complexity T is incorporated in the function, where $T/T_{\max}$ is used to normalize T , and the coefficient K is used to establish a link between the computational complexity and the prediction error. For example, setting $K = 0.02$ implies that when the prediction error is below 0.02° , the computational complexity will dominantly determine the outcome; else, the prediction error will be more dominant. Thus, the coefficient K can be seen as a weighting coefficient. Obviously, the larger the value of K is, the greater contribution the computational complexity will make to the penalty function. If computation time is not an issue, the coefficient K can be set to zero.

For the above-mentioned two points A and B, it is easy to see that the parameter set $P1$ yields smaller penalty function value at point A while the parameter set $P2$ yields smaller penalty function value at point B. Therefore, the process of determining the filter parameter set P is defined as the process of solving the following equation:

$$F(N, m) = \min. \quad (9)$$

Based on the penalty function concept, we designed the algorithm for an optimized finite memory filter as shown in Fig. 6.

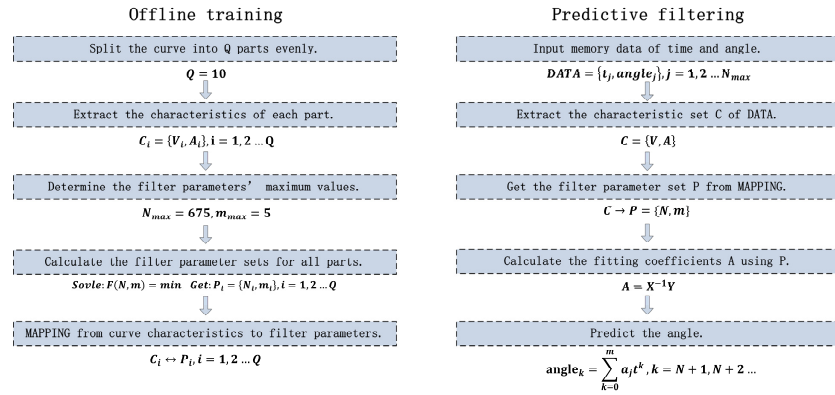


Fig. 6. An optimized finite memory filter.

Unlike the classical finite memory filter, the designed algorithm can be used for offline training and predictive filtering. Offline training is aimed at providing the optimal filter parameters for different pointing angle curves with different mobility; this training can be completed on the ground to ensure sufficient computational resources. The pointing angle curve can be obtained in advance by simulation or orbit prediction, and it will be split into several parts, following which the characteristics of each part will be extracted. The characteristics should represent the angular movement mobility, including the angular velocity V and the angular acceleration A . The angular velocity V can be directly provided by the ATP system, but the system in general does not have low-noise angular acceleration data. Thus, we used the curvature features of the angle curve to characterize the angular

acceleration A , and we obtained good results. Equation (11) is then separately solved for all curve parts by using the traversing method (it is also possible to use specially designed optimization algorithms). Finally, the mapping relationships between the curve characteristics and the filter parameters are obtained. The algorithm also implements real-time predictive filtering, which should be implemented in the ATP system to ensure the real-time computing ability. After the time code, the angle data (a combination of the measured angle and camera miss distance) and the velocity data are obtained, the memory curve characteristics are extracted, following which the corresponding filter parameters are determined by using the mapping relationships that were obtained during the offline training. Finally, the equations for the polynomial coefficients are solved to complete the prediction process.

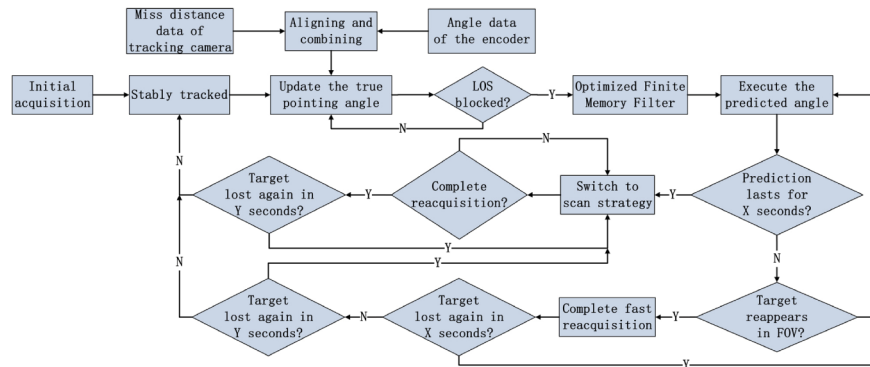


Fig. 7. Flow chart of fast reacquisition based on the optimized finite memory filter.

Figure 7 shows the flow chart of fast reacquisition including the optimized finite memory filter algorithm. After the initial acquisition, the target is stably tracked. During the tracking process, the ATP system automatically records a certain length (greater than or equal to $N_{\max}$) of miss distance data and measured angle data. The system then updates and stores the true pointing angle after aligning and combining the two data types. The camera's light spot centroid data is used to determine whether the LOS is blocked. If the LOS is not blocked, the above process is continued; otherwise, predictive filtering is performed by using the optimized finite memory filter, and open-loop pointing is executed according to the predicted angle. The prediction and pointing process will last for X seconds at most, where the value of X depends on the prediction performance and the system FOV. During this process, if the LOS is recovered and the prediction and pointing is accurate enough, the target will directly reappear in the FOV, then the search process can be shortened, and fast reacquisition is achieved.

However, if the prediction lasts for X seconds and fast reacquisition is not yet completed, the system should be switched to scan strategy to complete the reacquisition anyway. The scan strategy here might differ from the initial acquisition because it can still take advantage of the prediction results. In other words, the prediction error might still be smaller than the FOU, even though it exceeds the FOV. Thus, a specially designed scan strategy will possibly improve the reacquisition process further, but it will not be discussed in detail in this paper.

In real-life application scenarios, the ATP system may encounter intermittent blockade of LOS during dark periods, which will affect the valid data length of the filter, so we take some measures to ensure the prediction performance in this situation. As is shown in the flow chart, after a successful fast reacquisition, if the target light is lost again in X seconds from the beginning time of prediction, the system will go back to the latest valid prediction and continue to execute the predicted pointing angle. This method helps to take full use of the effective tracking data and the prediction results. In addition, to ensure a minimum valid data length for the filter, the system won't revert to the stably tracked status until Y seconds of stable tracking is made after reacquisition; otherwise, it would have to go for the scan process

for the lack of available prediction results. It should be noted that there are other ways to deal with this situation. For example, we can use non-uniform sampling during the intermittent blockade of LOS to keep the filter active all the time, which will make the algorithm more complex but can expand the effective time of the filter in a certain degree.

3.3 Simulation of the designed algorithm

The designed algorithm was implemented in MATLAB, and Trajectory 3 in Table 1 was used as the angle curve to be predicted. Mimicking the real conditions, the historical tracking data error was set to be $60\ \mu\text{rad}$ ($1\ \sigma$), and $5\ \mu\text{rad}$ ($1\ \sigma$) of white Gaussian noise was added to the ideal angle data and miss distance data to represent the measurement noise, while the miss distance data lagged one frame relative to the angle data. The simulation time interval of the algorithm was $0.02\ \text{s}$, and the temporal extent of the memory data was $13.5\ \text{s}$. Because link interruption may occur at any stage of a satellite pass over the ground station, the prediction time point was allowed to traverse the entire communication phase in the simulation, and the maximal predicted angle error in 5 seconds was calculated. Prediction error of the azimuthal and elevation angles at different prediction time points is shown in Fig. 8.

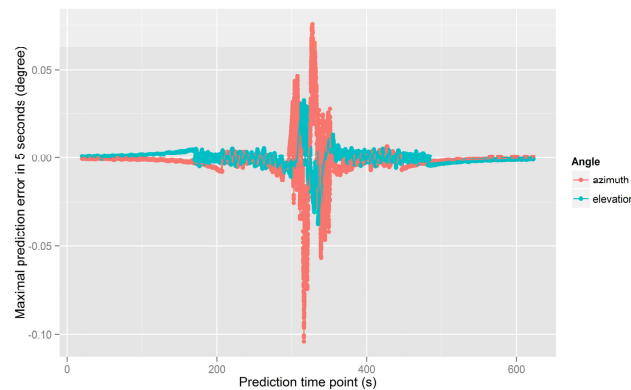


Fig. 8. Maximal prediction error in 5 seconds at different prediction time points using the optimized finite memory filter.

Figure 8 shows that, corresponding to the pointing angle curve regularity indicated in Fig. 3, the prediction error of the fast moving segment in the middle is greater than that at the both ends. As can be seen, compared to the finite memory filter with fixed parameters, the optimized algorithm effectively controls the prediction error at various stages by offline optimization of the filter parameters. The simulation results also indicate that well-trained filters exhibit good adaptability over different trajectories. Table 2 lists the angle prediction results for the three different trajectories shown in Table 1.

Table 2. Angle Prediction Results for Different Trajectories

Maximal prediction error in 5 seconds	Trajectory 1	Trajectory 2	Trajectory 3
Azimuth	0.006°	0.022°	0.08°
Elevation	0.005°	0.008°	0.03°

The root sum square value of the azimuthal and elevation angles error is used to evaluate the total prediction error. Taking Trajectory 3 as an example, the maximal prediction error of the pointing angle is calculated along with the prediction duration time. As Fig. 9 shows, the angle prediction error increases rapidly as the prediction time becomes longer; the total prediction error is mainly contributed by the azimuth for its stronger movement mobility.

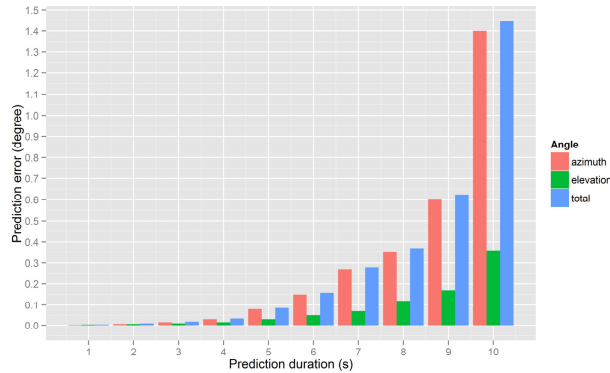


Fig. 9. Prediction error vs. the prediction duration.

In summary, due to the nonlinear and motion model unknown angle curve features, the prediction error is closely related to the angular movement mobility and prediction time. In addition, the prediction error increases with increasing high-frequency noise. Theoretically, using this algorithm it should be possible to obtain prediction accuracy higher than 0.1 degrees within 5 seconds.

4. Fast reacquisition experiment

4.1 Experimental setup

To test the performance of the designed algorithm in a real ATP system [25], an experimental system for acquisition was built in the laboratory by using a simulated dynamic target. Figure 10 shows the experimental system, which uses a dynamic target based on the wall reflection [26]. The two-axis target turntable drives the laser device, and the laser beam hits the wall and forms the reflection light spot, which can be used as the acquisition and tracking target of the ATP system. The two-axis target turntable is programmed according to the spatial relationship between the devices, to ensure the target angular movement relative to the ATP system meets the regularity of the satellite-to-ground pointing angle curve. Thus, the acquisition and the tracking processes of the space-borne ATP system over the ground station can be mimicked along with the reacquisition process in which the LOS is temporarily blocked and then recovers.



Fig. 10. Experimental system used for acquisition simulations in the laboratory.

The tracking and reacquisition FOV of the experimental ATP system is 0.2 degrees. To account for the tracking process interference by the diffuse reflection halos, an aperture screen is set in front of the tracking camera. Due to the limited computing resources of the ATP circuit box, the algorithm shown in Fig. 6 is implemented in the auxiliary computer, which communicates with the main control circuit of the ATP system in real-time via a serial port. The offline training of the algorithm is accomplished in advance in MATLAB; predictive filtering is incorporated into the ground-test software of the ATP system and is implemented in C++.

During the experiment, the ATP system acquires and tracks the dynamic target, and when the tracking status enters the dark period, prediction and pointing are executed according to the fast reacquisition process shown in Fig. 7, following which the actual prediction error and fast reacquisition performance are examined. For comparison, pointing and reacquisition should also be performed by using the classical data guiding methods. The guiding data for acquisition is often obtained by orbit prediction, and its position accuracy is generally on the order of one kilometer. With time offset effect and other factors added, the final FOU reaches ~ 0.25 degrees. Since the FOU is greater than the reacquisition FOV, scanning and searching are required when this acquisition method is used.

4.2 Experimental results and analysis

To assess the actual prediction accuracy of the algorithm, as the system exits the tracking status and enters the predicting and pointing status, the centroid position change of the dynamic target light spot in the ATP coarse tracking subsystem's camera FOV is checked. The farther the centroid position is from the FOV center, the larger prediction error is obtained. The result is shown in Fig. 11.

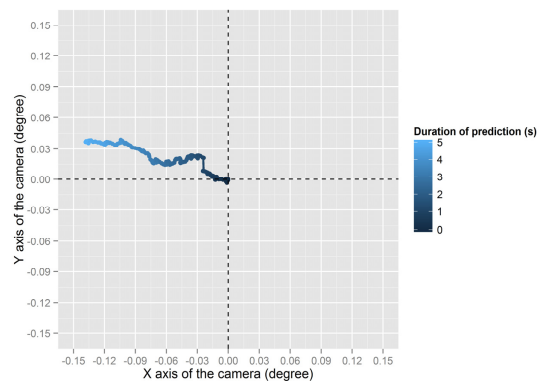


Fig. 11. Centroid position change of the light spot in the camera FOV during prediction.

In Fig. 11, the X-axis corresponds to the azimuthal direction and the Y-axis corresponds to the elevation direction. As can be seen from the figure, the prediction error increases with the prediction time, wherein the error in the X-axis direction is greater than that in the Y-axis, and the total prediction error within 5 seconds is smaller than 0.15° . Thus, it is conceivable that in an actual system it is possible to achieve prediction results that will be close to the simulation results. Besides, predictions obtained for different curve parts of the satellite-to-ground pointing angle are also consistent with the simulation results in Fig. 8. In other words, the fast moving segment near the zenith is characterized by a greater prediction error.

During the experiment, it was found that, in addition to the prediction algorithm itself, the execution error of the predicted angle also directly affected the final prediction accuracy. The execution error can be calculated by comparing the theoretically predicted angle with the actual output angle, and it consists of the pointing error and the output lag error. The former is a random error and is determined by the pointing accuracy of the actual system. The output lag error is a system error, which is caused by factors such as temporal delay of calculation

and execution, and temporal synchronization; this error can be corrected to a certain extent by adjusting the temporal offset of the extrapolation. The diagram of prediction error elements containing theoretical error and execution error is shown in Fig. 12, from which we can conclude that the execution error remains substantially the same over the time, and is significantly reduced after the timing correction, thereby improving the overall prediction accuracy.

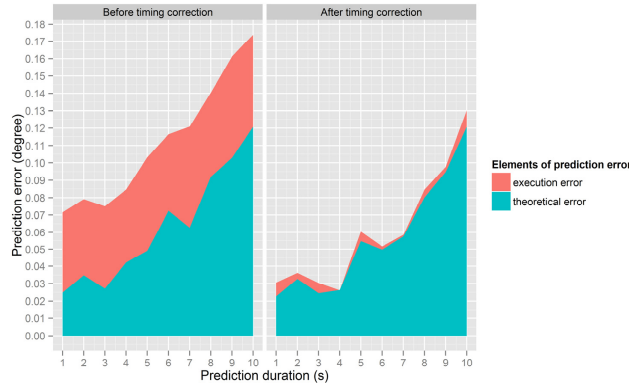


Fig. 12. Elements of prediction error and the effect of timing correction on the execution error reduction. (Left: Before timing correction; Right: After timing correction).

In the experiment, we ran more than ten trials with several different curves, whose angular velocity varied from $0.1^\circ/s$ to $0.5^\circ/s$. For different curves, the prediction results were different; but for the same curve, the prediction results were relatively stable in different trials. The longest prediction time that kept the target within the FOV varied from about 6 seconds to more than 15 seconds, and as long as the target did not get out of the FOV, it could be successfully and quickly reacquired. Figure 13 illustrates the actual reacquisition process with the camera centroid position change over time in a particular prediction and acquisition trial of the experiment.

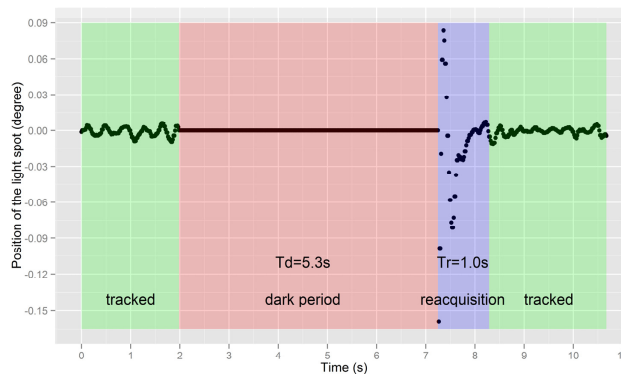


Fig. 13. Light spot centroid position change over time in a particular fast reacquisition process.

Figure 13 shows that, after a dark period of 5.3 seconds, since the prediction error is still smaller than the tracking FOV, the target immediately reappears in it, and after a short period of damped oscillation (which can be shortened by optimizing the controller parameters), the system completes fast reacquisition in 1 second. In contrast, if the classical acquisition method is used with the guiding data and spiral raster scanning pattern, the results of multiple tests indicate that the average reacquisition time is about ten seconds.

Thus, predictive filtering can effectively help achieve rapid reacquisition. However, according to the simulation and experimental results, for dark periods over 7 seconds the prediction accuracy will decrease to the similar level as the one obtained by using the guiding

data, and the golden time of fast reacquisition had passed, which implies classical orbit prediction should be used to complete the reacquisition. Besides, in this approach, the algorithm needs about 13.5 seconds of tracking data to keep the filter active.

5. Summary

In this paper, the validity of predictive filtering based on the angle information was analyzed, and an optimized prediction algorithm was designed based on the finite memory filter. Simulation results indicated good prediction performance of the algorithm, and laboratory acquisition experiments further demonstrated the effectiveness of the fast reacquisition approach in an actual ATP system. Achieving rapid reacquisition is important for improving the average data rate and reducing the beam power requirement in laser communication applications.

To further shorten the reacquisition time, more advanced design of reacquisition approach, such as combining the predictive filtering and scanning methods, should be studied in the future. And more experiments that simulate the real-life application scenarios should be conducted to validate the approach.

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